Roll No……..

Mid-Semester Examination 2025

Name of the Program: M.Tech. (Biomedical Engineering) *Semester: I*st
Name of the Course: Biomedical Electronics *Course Code:* BEM 103

Time: 90 minutes Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub-questions***.
(ii) Each question carries 10 marks

Q1	(10 marks)	**CO1**
(a)	Define physiological signals. Classify them with suitable examples.	
	OR	
(b)	Explain the physiology of nerve and muscle cells. How do action potentials arise?	
Q2	(10 marks)	**CO2**
(a)	Write notes on different types of bio-signals (electrical, mechanical, acoustic, chemical, optical, thermal) with examples.	
	OR	
(b)	Draw and explain the ECG waveform with its components (P wave, QRS complex, T wave).	
Q3	(10 marks)	**CO1**
(a)	Discuss the physiological signals obtained from the circulatory system.	
	OR	
(b)	What is the auditory system? Explain the generation and transmission of sound signals in the ear.	
Q4	(10 marks)	**CO2**
(a)	Compare ECG, PCG, and EMG in terms of origin, signal characteristics, and clinical use.	
	OR	
(b)	Discuss the significance of different brain waves (α, β, θ, δ) in EEG.	
Q5	(10 marks)	**CO1, CO2**
(a)	Describe the respiratory system and the physiological signals associated with respiration.	
	OR	
(b)	Explain the physiology of muscle contraction and how EMG is recorded	